

Statistical distributions, hypothesis testing,  
and creating actionable insights

# Why This Matters: From Raw Data to Real Questions

Before building complex predictive models, we must first understand our data and validate our assumptions.

In any field, moving from "I think" to "I know" is the key to success.

Data-driven decision-making is the process of using facts, metrics, and data to guide strategic business decisions that align with your goals.

## **Why It Matters:**

- Reduces Bias: It grounds your decisions in objective reality, not just intuition or anecdotal evidence.
- Increases Consistency: Creates a framework for making reliable choices time and time again.
- Uncovers Opportunities: Helps you spot trends, inefficiencies, and growth areas that would otherwise go unnoticed.

# Understanding Your Data's "Shape"

A statistical distribution is a function that shows the possible values for a variable and how often they occur.

Think of it as a "blueprint" for your data, describing its underlying pattern and probability.

They allow us to model real-world processes and understand variability.

Knowing the distribution of your data helps you:

- Identify expected outcomes and outliers.
- Make more accurate forecasts.
- Choose the right statistical tests for your analysis.

# Key Distribution Examples

## The Normal Distribution: The "Bell Curve"

A symmetric, bell-shaped curve where the mean, median, and mode are all the same.

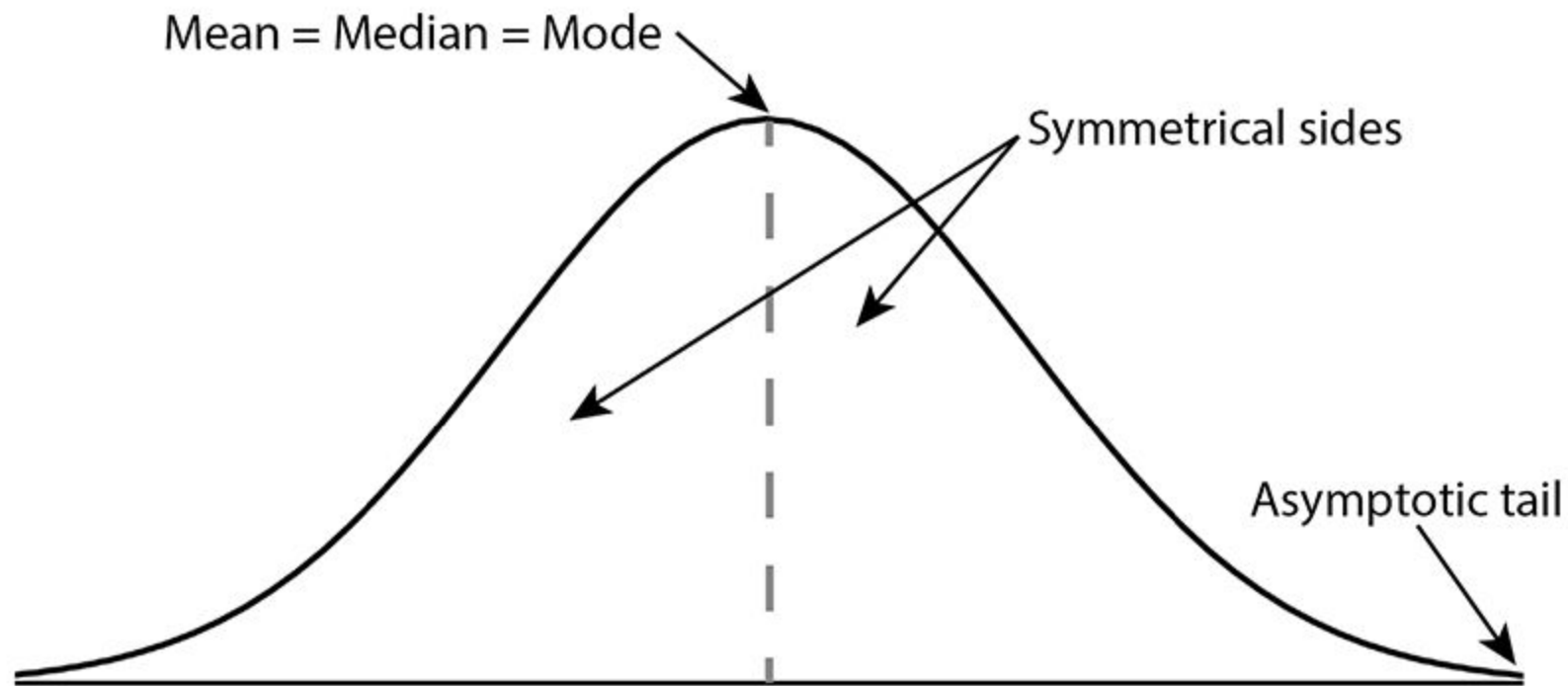
What it tells you: Most values cluster around the average. Values that are far from the average are increasingly rare. The data is predictable and balanced.

Examples:

- The distribution of customer satisfaction scores on a 1-10 scale.
- Measurement errors from a calibrated machine.
- The heights or blood pressure of a large population.

Why it's important: It's the foundation for many statistical tests (like the t-test). Many methods assume the data (or the errors in the data) are normally distributed.

# The Normal Distribution



# Cont.

## Skewed Distributions: The Reality of Financial Data

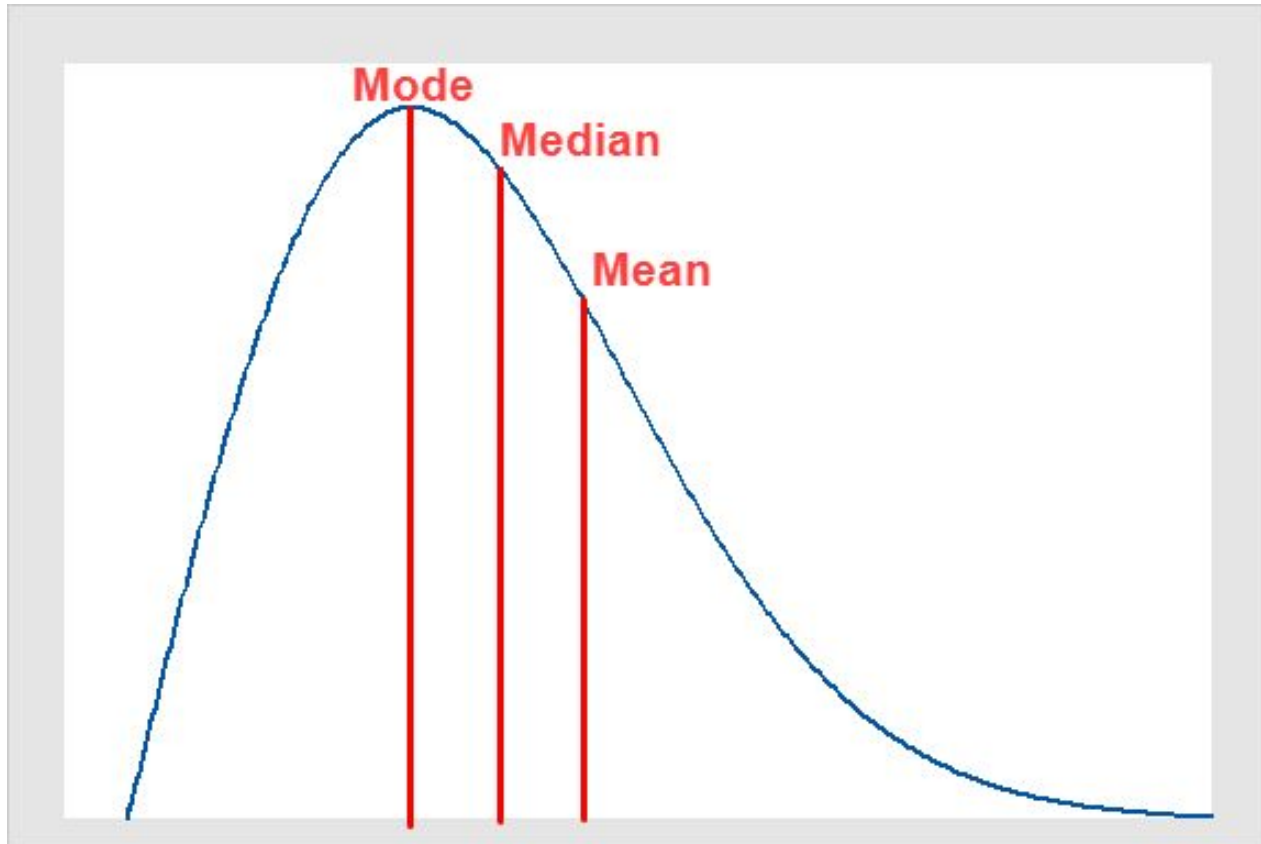
An asymmetric curve. The example below is Right-Skewed, where most values are clustered on the left, with a long tail of a few very high values to the right.

What it tells you: Most events are low-value, but there's a small chance of a very high-value event. In this case, the mean is pulled higher than the median.

Examples:

- Insurance Claim Amounts: Most claims are small, but a few are catastrophically large.
- Customer Income or Purchase Value: Most customers have modest incomes/purchases, but a few are extremely high.
- Time spent on a website: Most users leave quickly, but a few stay for a very long time.

Why it's important: The mean can be a misleading metric for skewed data. The median is often a better measure of the "typical" value. Ignoring skew can lead to a severe underestimation of risk.



# Cont.

## The Poisson Distribution: Modeling Event Counts

A bar chart of non-negative whole numbers. You can't have 2.5 events.

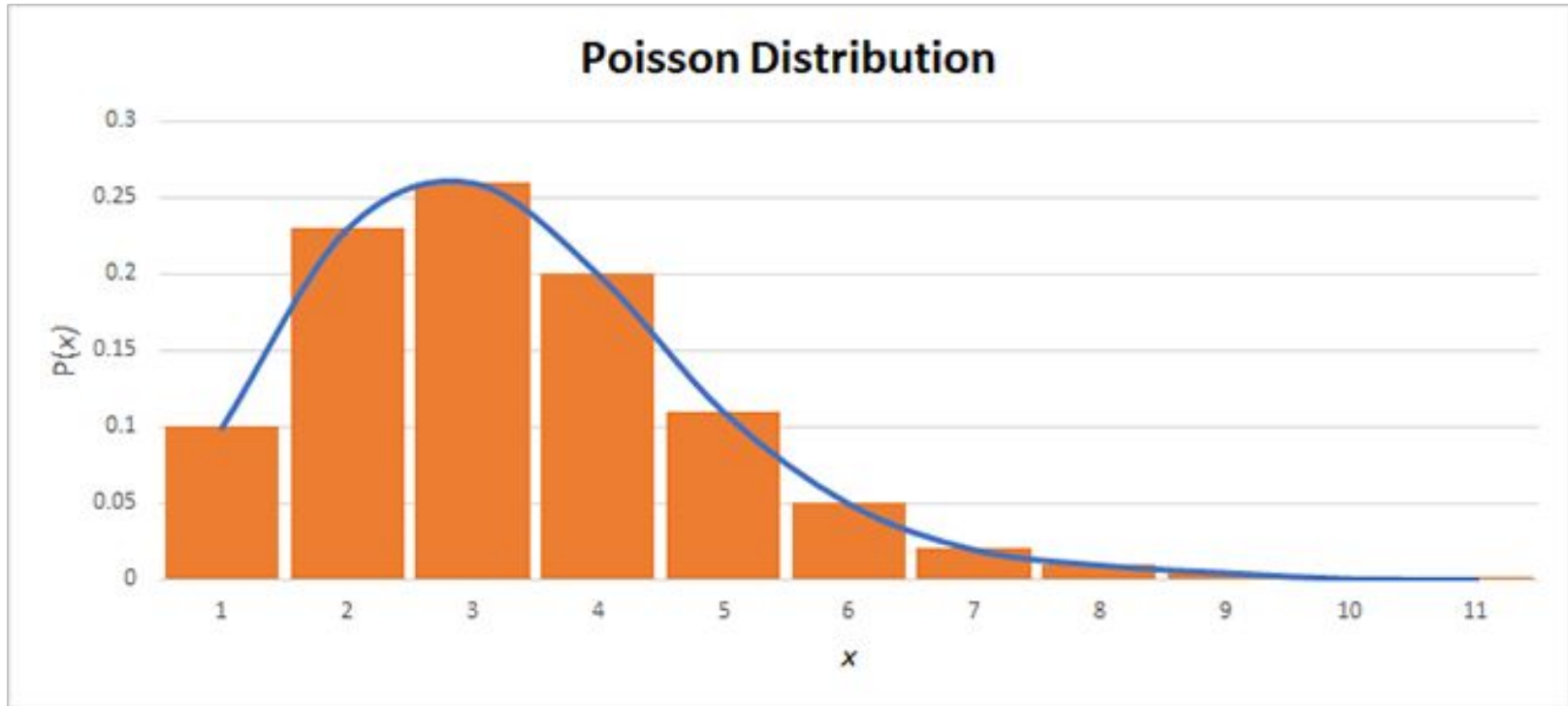
What it tells you: The probability of a given number of events happening in a fixed interval of time or space, given you know the average rate of occurrence ( $\lambda$ ).

Examples:

- Claim Frequency: The number of claims filed by a policyholder in a year.
- Operations: The number of customer support calls received in an hour.
- Marketing: The number of clicks on an ad per day.

Why it's important: It's the classic tool for modeling the frequency of events. This is essential for resource planning (e.g., staffing a call center) and for the "frequency-severity" approach to risk modeling.





# Asking and Answering Questions with Data

Hypothesis Testing is a formal procedure for checking if your data supports a particular theory. It helps you distinguish a real effect from random chance.

The Core Concepts:

Null Hypothesis ( $H_0$ ): The "default" or "no effect" assumption.

e.g., "There is no difference in average purchase value between Group A and Group B."

Alternative Hypothesis ( $H_1$ ): The theory you want to prove.

e.g., "There is a significant difference in average purchase value."

p-value: The probability of observing your data if the null hypothesis were true. A low p-value (typically  $< 0.05$ ) gives you a reason to reject the null hypothesis.

# Common Tool 1: The T-test

Purpose: To compare the average of a numerical variable between two different groups.

Business Question: "Does our new marketing campaign (Group B) generate a higher average revenue per user than the old campaign (Group A)?"

Hypotheses:

$H_0$ : The average revenue is the same for both campaigns ( $\mu_A = \mu_B$ ).

$H_1$ : The average revenue is different for the two campaigns ( $\mu_A \neq \mu_B$ ).

When to Use: When you have 1 numerical variable (e.g., TotalClaims) and 1 categorical variable with exactly 2 levels (e.g., Gender).

# Common Tool 2: ANOVA (Analysis of Variance)

Purpose: An extension of the t-test to compare the averages of a numerical variable across three or more groups.

Business Question: "Do different vehicle types (e.g., Sedan, SUV, Truck) have different average claim amounts?"

Hypotheses:

$H_0$ : The average claim amount is the same for all vehicle types ( $\mu_{\text{Sedan}} = \mu_{\text{SUV}} = \mu_{\text{Truck}}$ ).

$H_1$ : At least one vehicle type has a different average claim amount.

When to Use: When you have 1 numerical variable (e.g., TotalClaims) and 1 categorical variable with 3 or more levels (e.g., Province, VehicleType).

# Common Tool 3: The Chi-Squared ( $\chi^2$ ) Test

Purpose: To check for a relationship between two categorical variables. It compares observed frequencies to expected frequencies.

Business Question: "Is there a relationship between a customer's gender and the type of vehicle they insure (e.g., Sedan vs. SUV)?"

Hypotheses:

$H_0$ : Gender and Vehicle Type are independent (there is no association).

$H_1$ : Gender and Vehicle Type are dependent (there is an association).

When to Use: When you have 2 categorical variables (e.g., Gender and VehicleType).

# The Bridge from Data to Decisions

A p-value tells you if an effect is likely real or just random noise. It does not tell you how big the effect is or what to do about it. The real value is created in the steps that follow.

We can use a simple but powerful framework:

What? - State the statistical finding clearly and concisely.

So What? - Interpret the finding. What does it mean in a business context? How big is the effect?

Now What? - Recommend a specific, data-driven action or next step.

# Example 1: Insight from an ANOVA Test

Scenario: You used ANOVA to test if the average claim amount differs across three provinces and found a significant result.

What?

"The ANOVA test was statistically significant ( $p < 0.01$ ), rejecting the null hypothesis that average claim amounts are equal across all provinces."

So What?

"Post-hoc analysis shows the average claim in Province B is 22% higher than in Provinces A and C. This means our financial risk exposure is not uniform; we are currently more vulnerable to larger losses in Province B, which negatively impacts our profitability in that region."

Now What?

"I recommend we initiate a targeted investigation into the underlying risk factors specific to Province B (e.g., higher vehicle theft rates, road conditions). Based on the findings, we should then model a potential premium adjustment for this specific region."

## Example 2: Insight from a Chi-Squared Test

Scenario: You used a Chi-Squared test to see if there's a relationship between having an anti-theft device and having a claim.

What?

"The Chi-Squared test was statistically significant ( $p < 0.05$ ), indicating a strong association between having an anti-theft device and claim status."

So What?

"Customers with a certified anti-theft device are 40% less likely to file a claim compared to those without one. This feature is a powerful indicator of a lower-risk customer segment."

Now What?

"I recommend we introduce a new premium discount for policyholders who can prove they have a certified anti-theft device installed. This will attract and retain lower-risk customers, directly improving our overall loss ratio."



# Common Pitfalls to Avoid

Translating results requires careful thought. Here are two common traps:

Confusing Statistical vs. Business Significance:

The Trap: A result can be statistically significant ( $p = 0.01$ ) but have a tiny effect size (e.g., a \$1 difference in average claims).

The Fix: Always ask, "Is this difference large enough to matter to the business?" Focus your energy on findings that are both statistically significant and have a meaningful impact.

Correlation is Not Causation:

The Trap: Your analysis shows that customers who own red cars have higher claim rates. You conclude that the color red causes risky driving.

The Fix: Acknowledge that there might be a hidden (lurking) variable. Perhaps people who prefer red cars also tend to prefer sports cars, which are inherently riskier. Your insight should be: "We've found a correlation. Our next step is to investigate the potential root causes before making any decisions."

# Summary & Your Analytical Starting Point

Start with Distributions: Plot your key variables. Understand their shape, center, and spread. Are they skewed? This is your first insight!

Formulate a Hypothesis: Turn a business question into a testable null hypothesis based on the variables in your data.

Choose the Right Test:

- Comparing 2 group averages? T-test.
- Comparing 3+ group averages? ANOVA.
- Comparing 2 categorical variables? Chi-Squared Test.

Translate Your Result: Don't just report the p-value. Explain what it means for the business and recommend a next step. This is how you create value.

Any questions?